

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Enabling agents to autonomously regulate
knowledge utilization through situational self-
awareness.

Source: arxiv.org/abs/2504.03553

Current Agent Training Uses 'Flood Irrigation'

- The Problem: Indiscriminate Knowledge Injection
 - Existing approaches blindly inject gold trajectories, external feedback, and domain knowledge without considering situational needs.
- Three existing paradigms & their limitations:
 - 1. Direct Trajectory Imitation (e.g., ReAct): Pattern-fitting without deeper understanding.
 - 2. Trial-and-Error Refinement (e.g., Reflexion): Inefficient and doesn't know when to seek external help.

Three Situational Modes of Thinking

Agent Decision Step
(Evaluate Current Situation)



Fast Thinking

Criterion: Agent can directly produce correct action.
Result: Direct Action



Slow Thinking

Criterion: Initial error, but correctable via reflection.
Result: Pause & Reflect



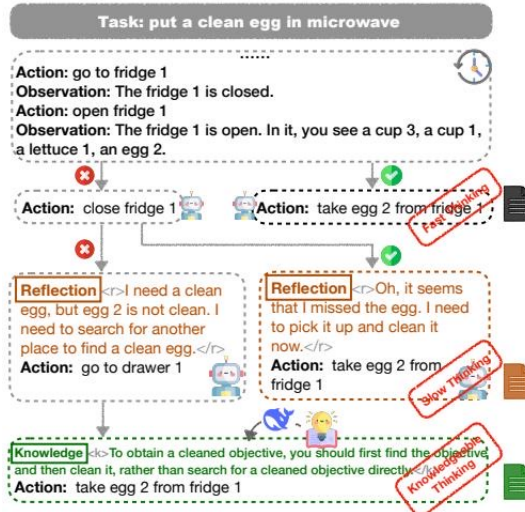
Knowledgeable Thinking

Criterion: Fails even after reflection.
Result: Consult Knowledge Base

KnowSelf Framework: Pipeline Overview

Step 1

Self-awareness Data Construction



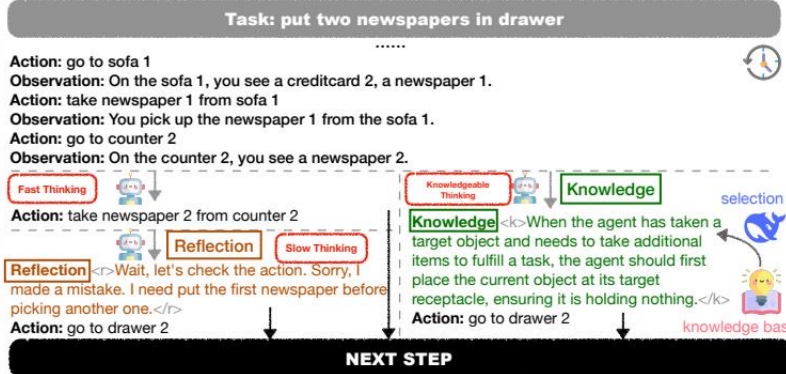
Step 2

Self-awareness Learning



Step 3

Self-awareness Inference



Three-Step Pipeline:

1. Data Construction: Heuristic situation judgment marks special tokens on self-explored trajectories.
2. Two-Stage Training: Supervised Fine-Tuning (SFT) followed by Reward-based Policy Optimization (RPO).
3. Self-Aware Inference: Selective token generation triggers reflection or knowledge retrieval only when needed.

Qualitative Examples: Superior Problem Solving

(a) Task: put a hot mug in cabinet	
OpenAI-O1 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge] <k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection] <r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer	
DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection] <r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge] <k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> Action: put saltshaker 1 in/on drawer 1

Task 1: Heat a Mug

KnowSelf uses reflection to heat the mug while OpenAI-O1 fails by trying to take an egg.

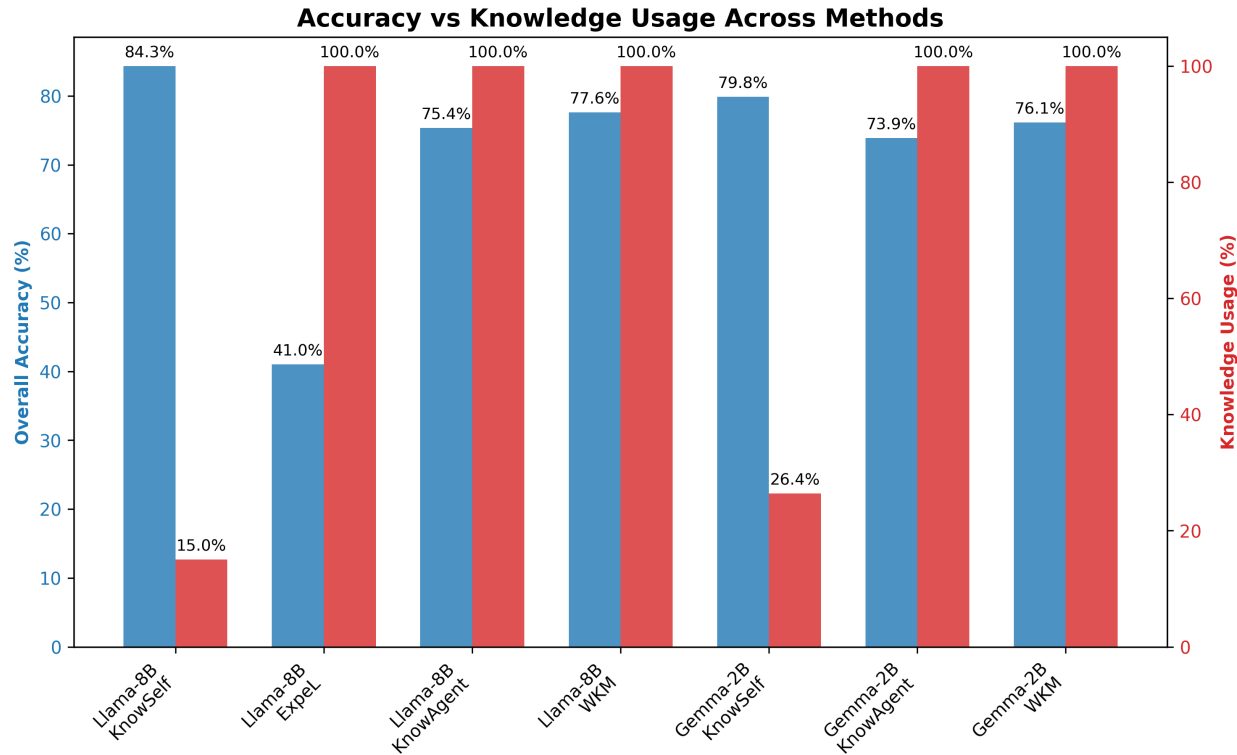
Task 2: Place a Saltshaker

KnowSelf applies knowledge to place the object without moving unrelated items, while DeepSeek-R1 gets stuck interacting with an open drawer.

Main Results on ALFWorld

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.0	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Efficiency: Minimal Knowledge, Maximum Impact



Key Observations:

Selective Advantage:

KnowSelf uses only ~15% (Llama-8B) to 26% (Gemma-2B) knowledge queries.

Outperforming Full-Knowledge Methods:

Methods like KnowAgent, WKM, and Expel utilize 100% knowledge but underperform KnowSelf. This confirms that indiscriminate knowledge injection can degrade planning.

Scalability:

Even a small Gemma-2B model paired with KnowSelf outpaces the 100% knowledge Gemma-2B baselines.

Key Takeaways: Self-Awareness is the Missing Piece

- 1. Indiscriminate knowledge injection is wasteful and suboptimal.
- 2. Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling.
- 3. KnowSelf with Llama-8B outperforms strong baselines including GPT-4o ReAct (84.33% vs 76.12%) and GPT-4o ExpeL (79.85%).
- 4. Only 15-26% knowledge usage needed versus 100% for competing methods.
- 5. The paradigm generalizes across model scales — even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL.